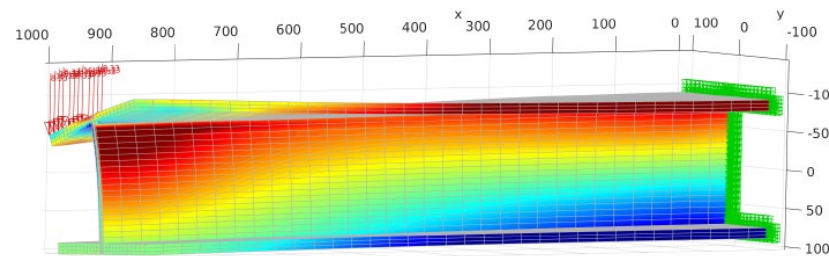


# Stochastic Finite Element Methods

## #1: Foundations of Uncertainty Quantification



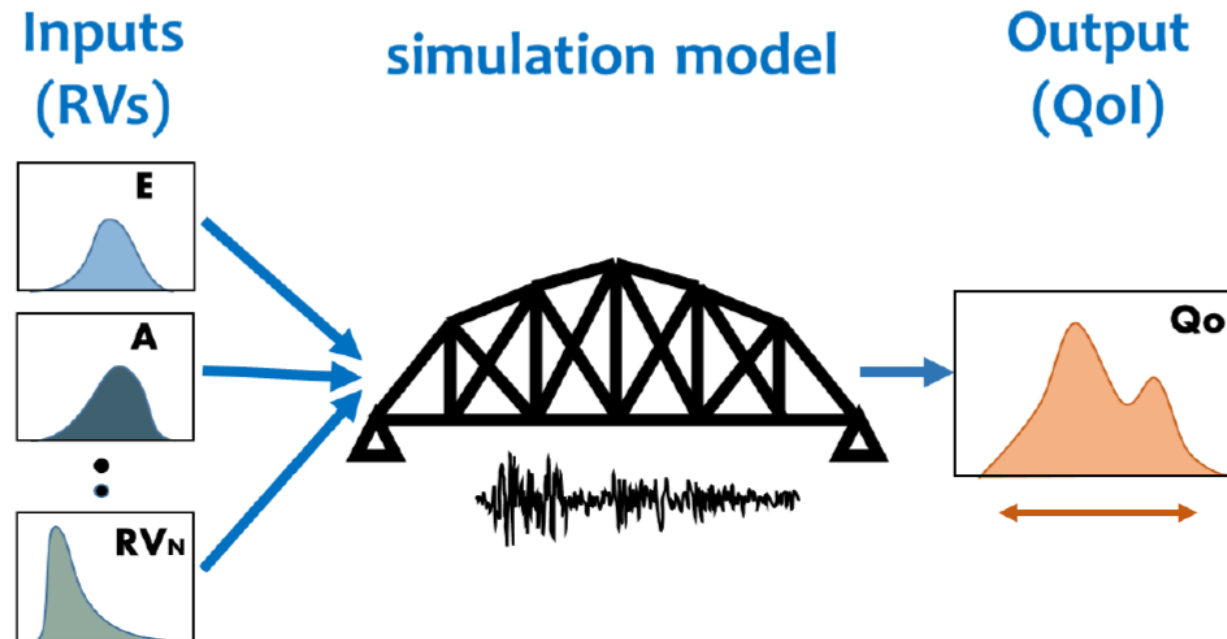
Prof. Dr.-Ing. Udo Nackenhorst  
Dr. Zhibao Zheng  
IBNM – Leibniz Universität Hannover

# Motivation

- Deterministic Finite Element Analysis is well established in engineering
- However, model parameters, e.g. material properties, boundary conditions and mechanical excitation are often or in general uncertain
- Mathematical descriptions and computational techniques for the treatment of stochastic PDE's are discussed with increasing intensity in scientific literature since approx. 2 decades, however, so far sound mathematical approaches are not established in engineering
- Goal of this classes is on the introduction of stochastic Finite Element Analysis for problems in solid body mechanics, where different solution techniques will be analyzed with regard to the applicability to engineering problems

# Motivation

- Why do we need to account for uncertainty in FEM simulations?



- This is more a class Uncertainty Quantification/Propagation than on FEM
- The course is on the intersect between statistical methods and machine learning

# The Core Problem: Reality is Uncertain

## Deterministic FEM assumes:

- Exact material parameters ( $E$ ,  $\nu$ ,  $\rho$  ...)
- Precisely known geometry
- Deterministic boundary conditions
- No measurement or model error

→ **In reality, none of this holds exactly!**

## Engineering Reality:

- **Material scatter**  $\pm 10\text{--}30\%$  CV in composites
- **Geometric tolerances** Manufacturing variability
- **Load uncertainty** Wind, traffic, seismic loads
- **Model uncertainty** Constitutive law errors
- **Measurement noise** Sensor and data errors

# Why it Matters: Engineering Consequences



## Safety & Reliability

Structural failure probabilities must be quantified. Deterministic safety factors are insufficient for life-critical components.



## Design Optimisation

Robust design requires accounting for scatter in inputs. Optimal designs ignoring uncertainty may be fragile.



## Model Validation

Experimental data shows scatter. Comparing deterministic predictions to data without UQ is statistically meaningless.



## Decision Under Risk

Risk-informed engineering decisions (RBDO, SHM) explicitly require probabilistic output quantities.

# Probability Spaces

## The Probability Triple $(\Omega, \mathcal{F}, \mathbb{P})$

$\Omega$

### Sample Space

- Set of all possible outcomes  $\omega$
- Example:  $\Omega = \mathbb{R}$  (Young's modulus values)
- Must be well-defined and exhaustive

$\mathcal{F}$

### $\sigma$ -Algebra

- Collection of measurable events  $A \subseteq \Omega$
- Closed under complement & countable unions
- Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R})$  is the standard choice

$\mathbb{P}$

### Probability Measure

- $\mathbb{P} : \mathcal{F} \rightarrow [0,1]$  (normalised measure)
- $\mathbb{P}(\Omega) = 1, \mathbb{P}(\emptyset) = 0$
- $\sigma$ -additivity for disjoint events

- » Measure-theoretic foundations underpinning stochastic analysis
- » *The probability triple  $(\Omega, \mathcal{F}, \mathbb{P})$  is the rigorous foundation for all random quantities in SFEM.*

# Basics of Statistics

Random Variable  $X : (\Theta, \mathcal{F}, \mathcal{P}) \rightarrow \mathbb{R}$

Probability Space  $(\Theta, \mathcal{F}, \mathcal{P})$

Sample Space  $\Theta$  event  $\theta \in \Theta$

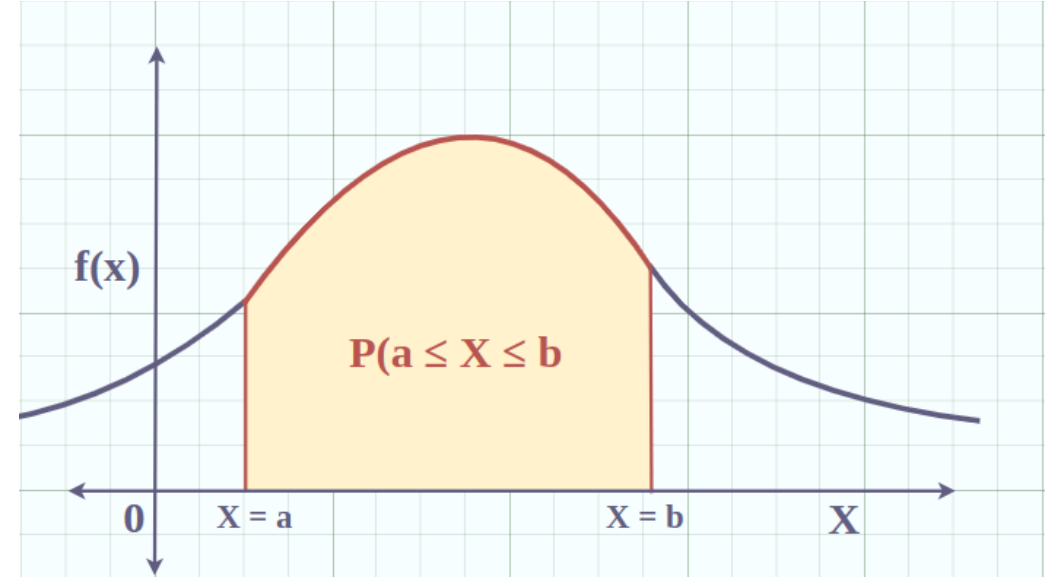
$\sigma$ -algebra  $\mathcal{F}$

Probability of an event  $\mathcal{P}(\theta)$

Probability density function  $\mathcal{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$

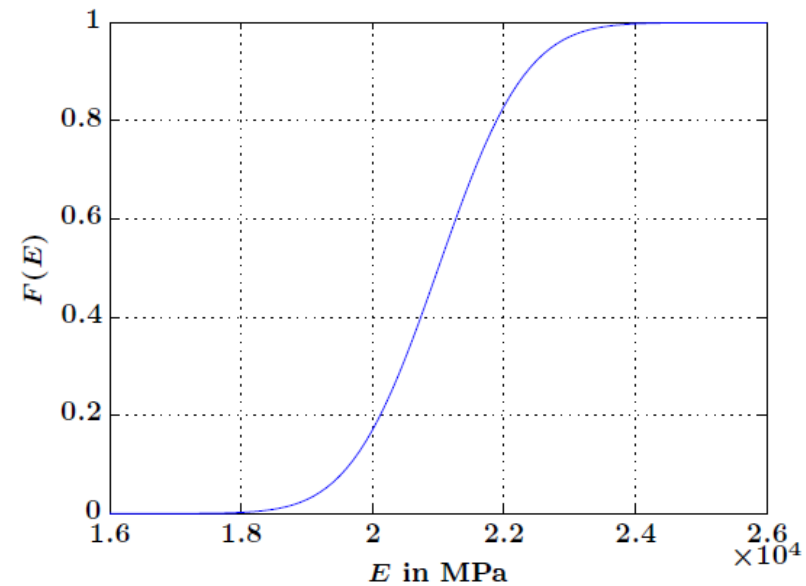
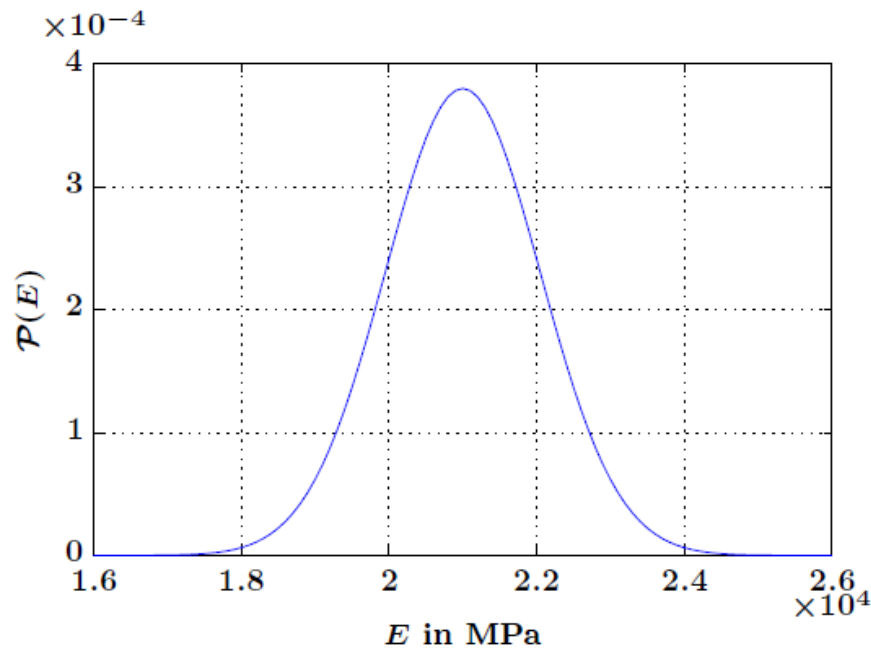
with  $f_X(x) \geq 0$  and  $\int_{-\infty}^{\infty} f_X(x) dx = 1$

Cumulative distribution function  $f_X(x) = \mathcal{P}(X \leq x)$



# Example

Gaussian distribution of Young's-modulus  $f_E(x) = \frac{1}{\sqrt{2\pi\sigma_E^2}} e^{-\frac{(x-\mu_E)^2}{2\sigma_E^2}}$



Mean value  $\mu_E = 210\text{GPa}$

Standard deviation  $\sigma_E = 5\%$



# Statistical Moments

In general:  $\mathbb{E}\{X^n\} = \int_{-\infty}^{\infty} x^n f_X(x) \, dx$

mean:  $\mu_X := \mathbb{E}\{X\} = \int_{-\infty}^{\infty} x f_X(x) \, dx$

variance:  $\sigma_X^2 = \text{var}(X) = \mathbb{E}[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) \, dx$

$$= \int_{-\infty}^{\infty} x^2 f_X(x) \, dx - \mu_X^2$$

skewness  $s_X = \frac{\mathbb{E}[(X - \mu_X)^3]}{\sigma_X^3}$

kurtosis  $\kappa_X = \frac{\mathbb{E}[(X - \mu_X)^4]}{\sigma_X^4}$

# More than one Random Variable

Joint probability density  $f_{XY}(x, y) \geq 0$

such that  $\mathcal{P}\{(X, Y) \in \mathcal{B}\} = \int_{\mathcal{B}} f_{XY} \, dx \, dy$

Marginal probability density functions:

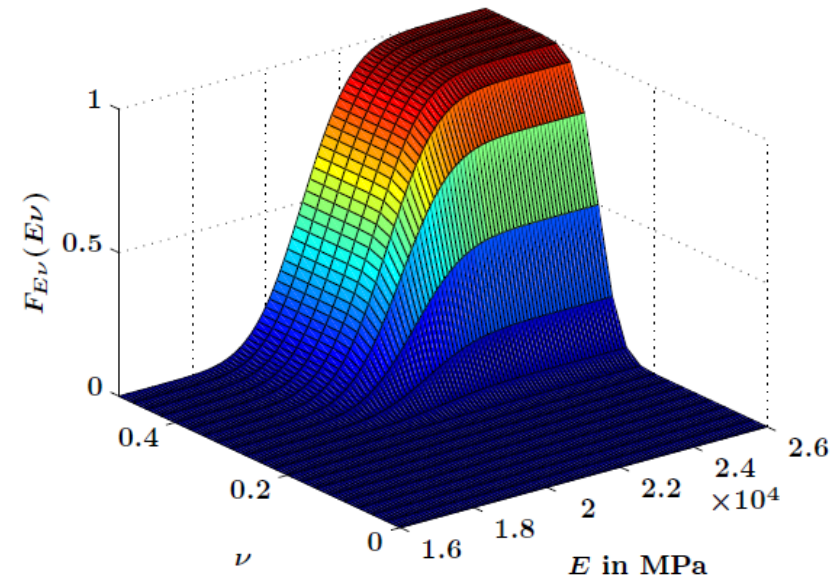
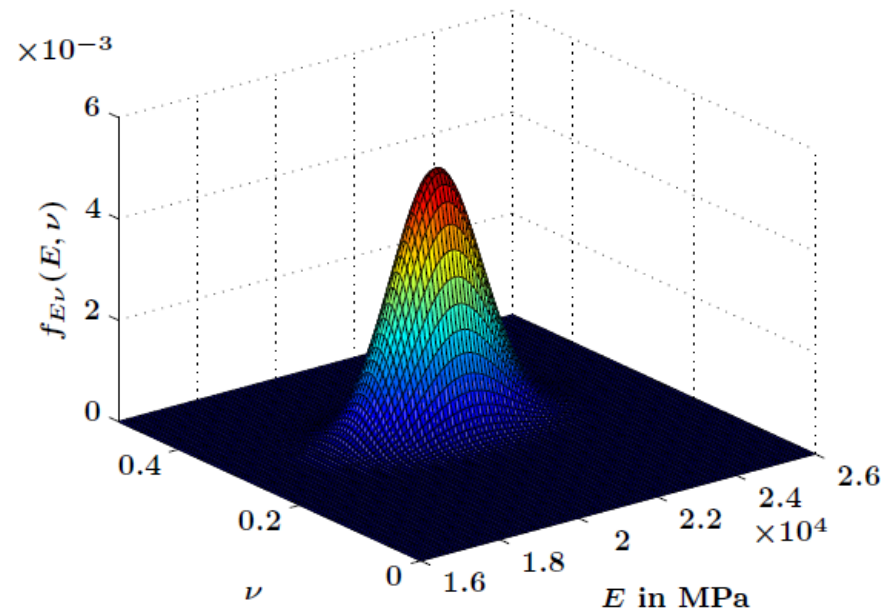
$$f_X(x) = \int_{\mathcal{B}_Y} f_{XY} \, dy \quad \text{and} \quad f_Y(y) = \int_{\mathcal{B}_X} f_{XY} \, dx$$

Joint cumulative density function

$$F_{XY}(a, b) = \mathcal{P}\{X \leq a, Y \leq b\} = \int_{-\infty}^a \int_{-\infty}^b f_{XY} \, dx \, dy$$

# Example

Gaussian distribution of Young's modulus and Poisson Ratio



$$\mu_E = 210\text{GPa} \quad , \quad \sigma_E = 5\%$$

$$\mu_\nu = 0.3 \quad , \quad \sigma_\nu = 10\%$$

# Correlation between Random Variables

Covariance  $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$   
 $= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$

Independent Variables:  $\text{Cov}(X, Y) = 0$

Random Vectors:  $\mathbf{X}^T = [X_1, X_2, \dots, X_n]$

Covariance Matrix:

$$\Sigma = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Cov}(X_n, X_n) \end{bmatrix}$$

# Conditional Probability

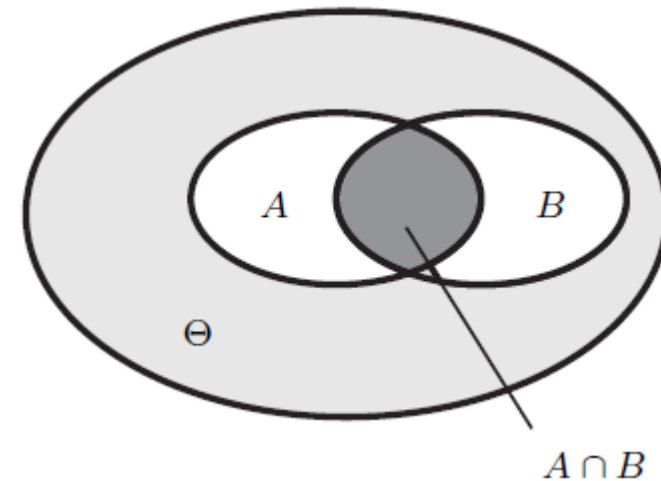
$$\mathcal{P}(A|B) = \frac{\mathcal{P}(A \cap B)}{\mathcal{P}(B)}$$

A and B independent:

$$\mathcal{P}(A \cap B) = \mathcal{P}(A)\mathcal{P}(B)$$

thus:

$$\mathcal{P}(A|B) = \mathcal{P}(A)$$



# Random Fields

Let  $(\Theta, \mathcal{F}, \mathcal{P})$  be the probability space and  $\mathcal{L}^p(\Theta, \mathcal{F}, \mathcal{P})$  a space, which is a linear normed space of a set of RV.

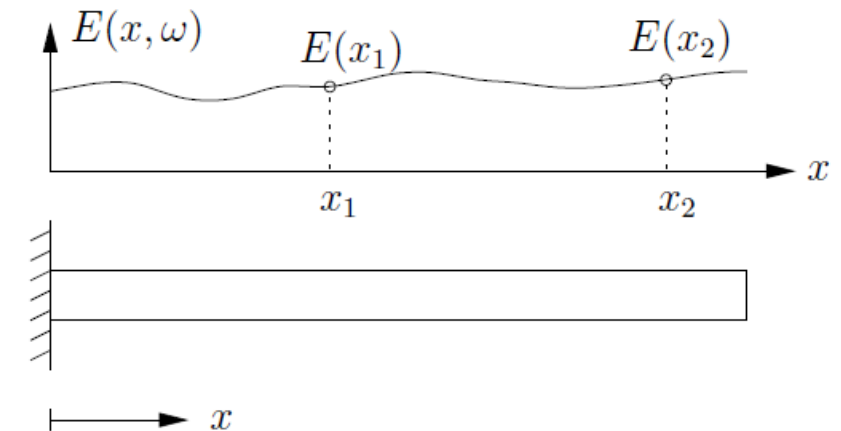
A random field  $H(x) \in \mathcal{L}^p(\Theta, \mathcal{F}, \mathcal{P})$  is a set of random variables indexed by the continuous variable  $x$ :

$$H : \mathcal{D} \times \Theta \rightarrow \Omega \subset \mathbb{R}^n$$

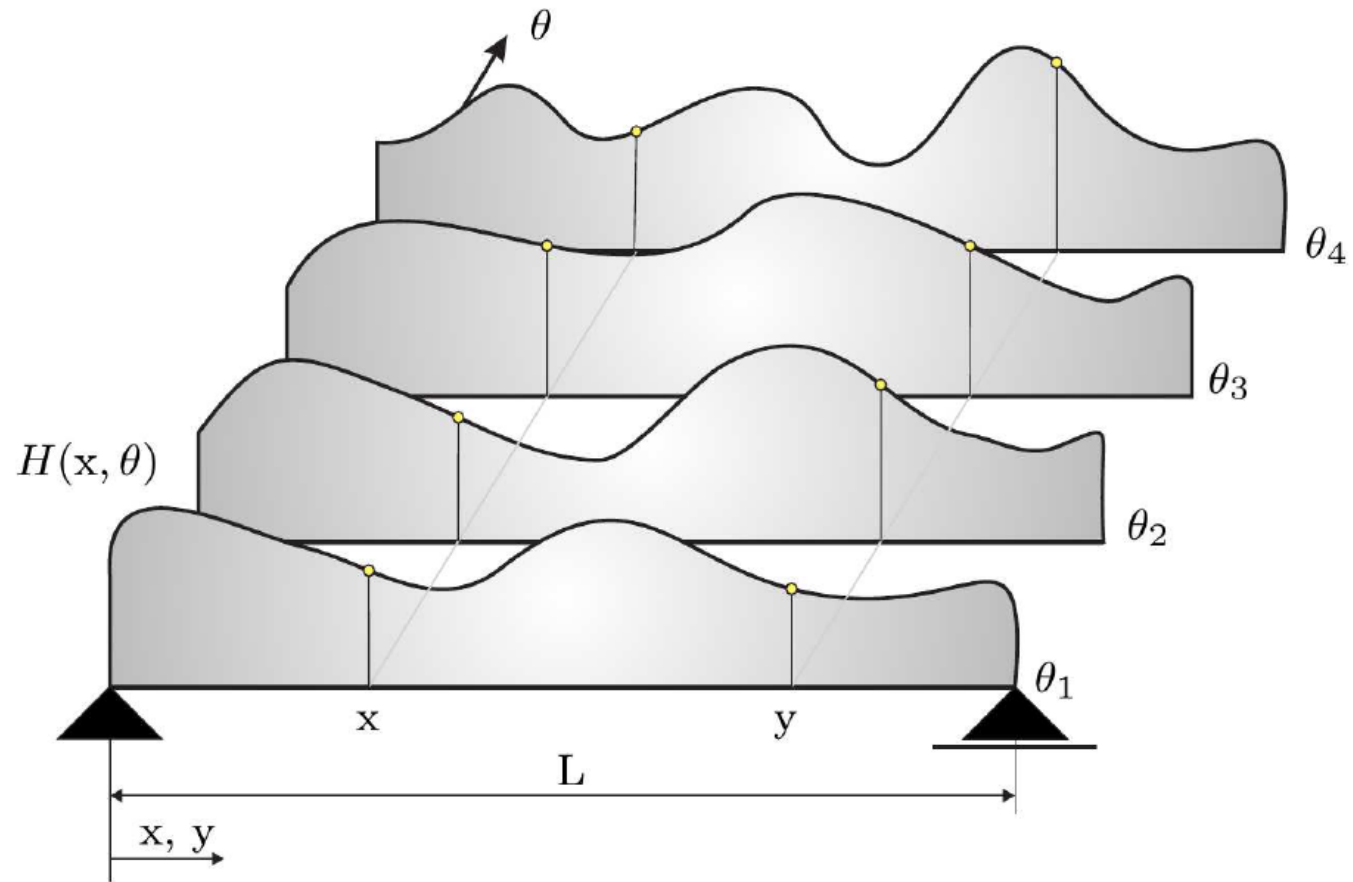
$$(x, \theta) \mapsto H(x, \theta)$$

- For a given  $\theta_0$   
 $H(x, \theta_0)$  is a realization of the random field.
- For a given  $x_0$   
 $H(x_0, \theta)$  is a random variable.

$\Omega$  is the state space of the process



# Realisations of a one-dimensional RF



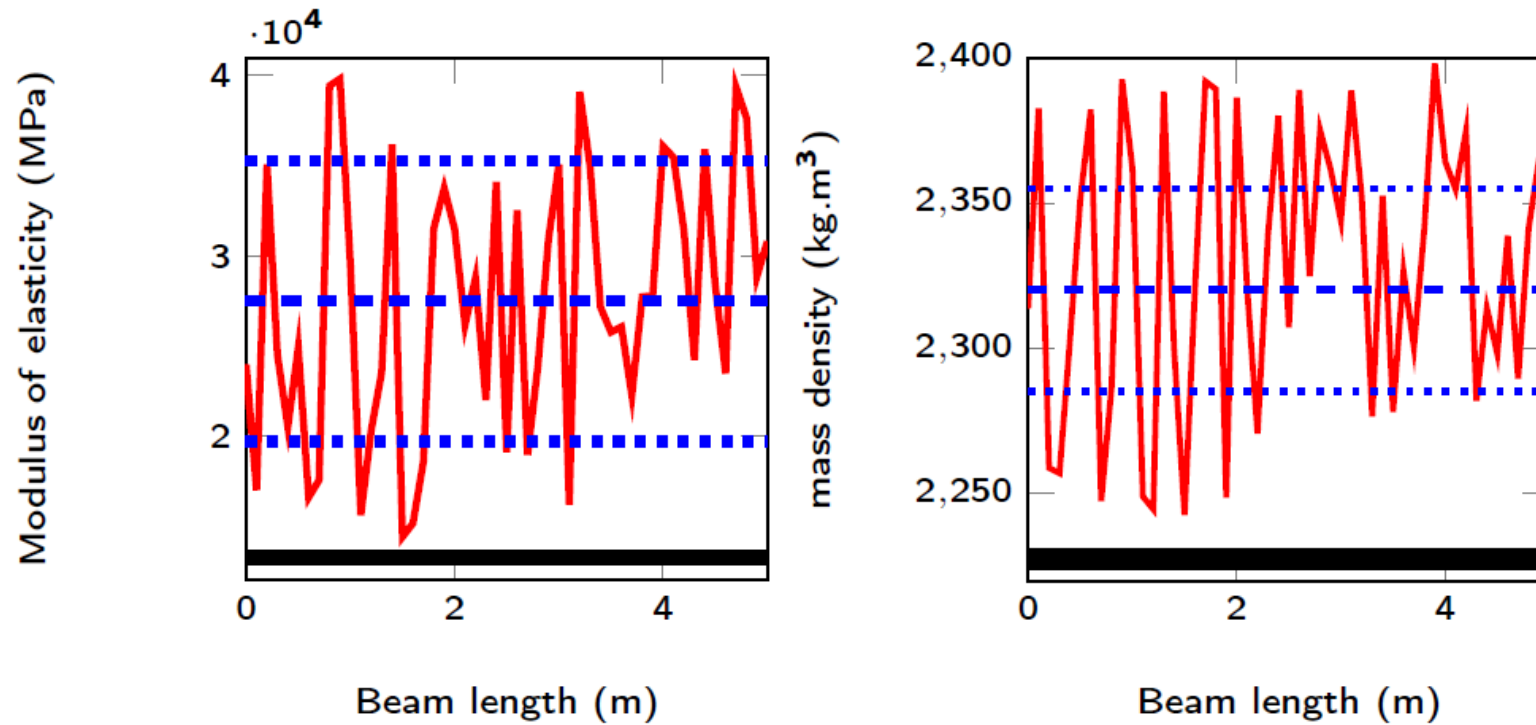
# Random Process

- $n$ -dimensional when  $x$  has the dimension  $n$
- Multivariate when the random variable  $H(x_0, \theta)$  has a dimension  $N > 1$
- Univariate when the random variable  $H(x_0, \theta)$  has a dimension equal to 1.
- In particular we call a random process a random field for which  $\mathcal{D}$  is the Euclidean space.



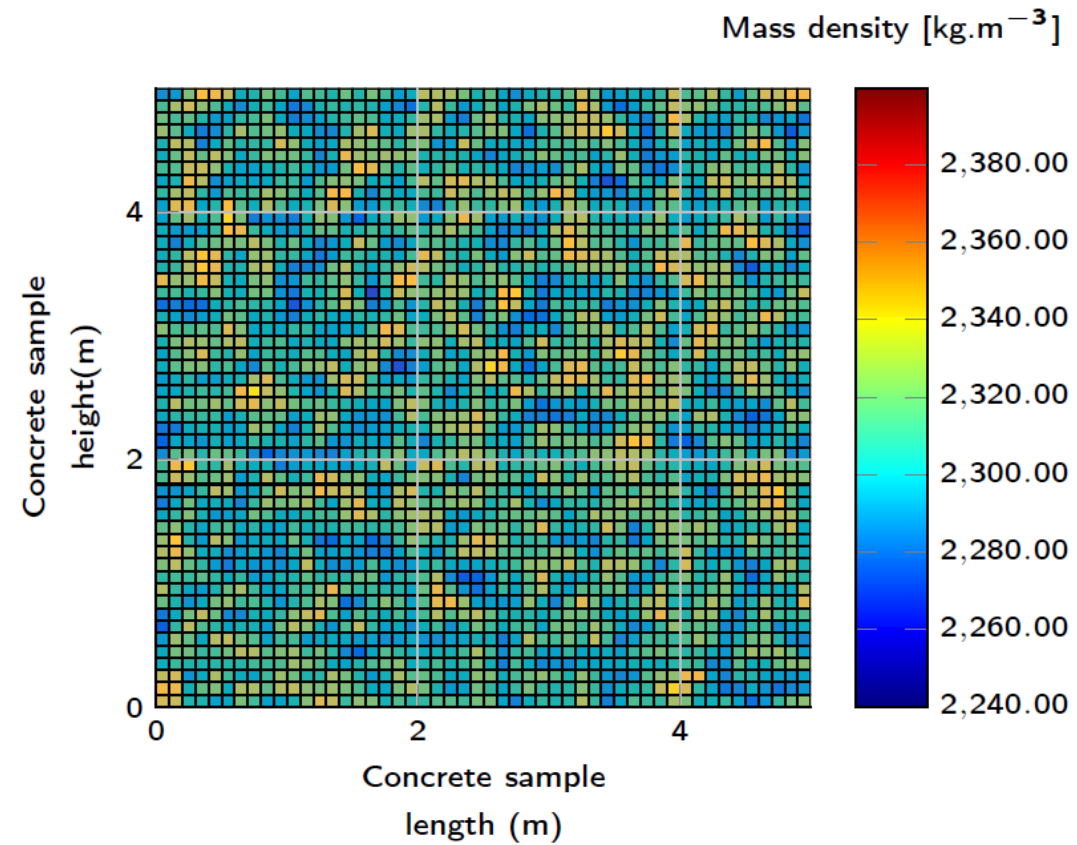
# Random Fields

Example of a two-variate one-dimensional RF



# Random Fields

Example of a univariate two-dimensional RF



# Gaussian Process

A Gaussian process has a joint probability density function

$$f_n(x_1, t_1; \dots x_n, t_n) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma_{xx})}} e^{-\frac{1}{2}(x - \mu_x(t))^T \Sigma_{xx}^{-1}(t)(x - \mu_x(t))}$$

Where  $\mu_x = \mathbb{E}\{H(t)\}$  is the expectation,  $\Sigma_{xx}(t)$  is the covariance matrix and  $x = x_i e_i$  is a vector in space.

- A multi-dimensional and multi-variate Gaussian process is fully described by its mean, second moment and covariance matrix.
- Many variables are empirically well approximated by Gaussian due to the central limit theorem.  
Nonetheless, Gaussian distributions are defined from  $-\infty$  to  $+\infty$ .  
Physical properties may not be negative!

# Probability Distribution of a RF

- A  $n$ -dimensional cumulative density function:

$$F_n(x_1, \dots, x_n) = \mathcal{P}(H(X_1, \theta) \leq x_1, \dots, H(X_n, \theta) \leq x_n).$$

- Probability density function:

$$f_n(x_1, \dots, x_n) = \frac{\partial^n F_n}{\partial x_1 \cdots \partial x_n}$$

- Probability density functions  $f_m$  relative to a number  $m < n$  of variables

$$f_m(x_1, \dots, x_m) = \int_{\mathcal{D}^{n-m}} f_n(x_1, \dots, x_n) \, dx_{m+1} \cdots dx_n$$

# Moments of a Random Process

- $m$ -th moment of a stochastic process  $X(t)$

$$\mu_x^m(t) = \mathbb{E}\{X^m\} = \int_{\mathcal{D}} x^m f_1(x, t) \, dx$$

- $m$ -th central moment of a stochastic process  $X(t)$

$$\bar{\mu}_x^m(t) = \mathbb{E}\{(X - \mu_x)^m\} = \int_{\mathcal{D}} (x - \mu_x)^m f_1(x, t) \, dx$$

- $mn$ -th joint moment at the given times  $t_1$  and  $t_2$

$$\begin{aligned} \mu_x^{mn}(t_1, t_2) &= \mathbb{E}\{(X^m(t_1)X^n(t_2))\} \\ &= \int_{\mathcal{D}^m} \int_{\mathcal{D}^n} x_1^m x_2^n f_2(x_1, t_1; x_2, t_2) \, dx_1 \, dx_2 \end{aligned}$$

# Autocovariance Function

- Description of the linear dependency of the random field for couples of points  $(x, y) \in \mathcal{D}^2$

$$C_{HH} : \mathcal{D}^2 \rightarrow \mathbb{R}$$

$$(x, y) \mapsto \mathbb{E}\{(H(x) - \mathbb{E}\{H(x)\})(H(y) - \mathbb{E}\{H(y)\})\}$$

- or considering two time instants  $t_1$  and  $t_2$

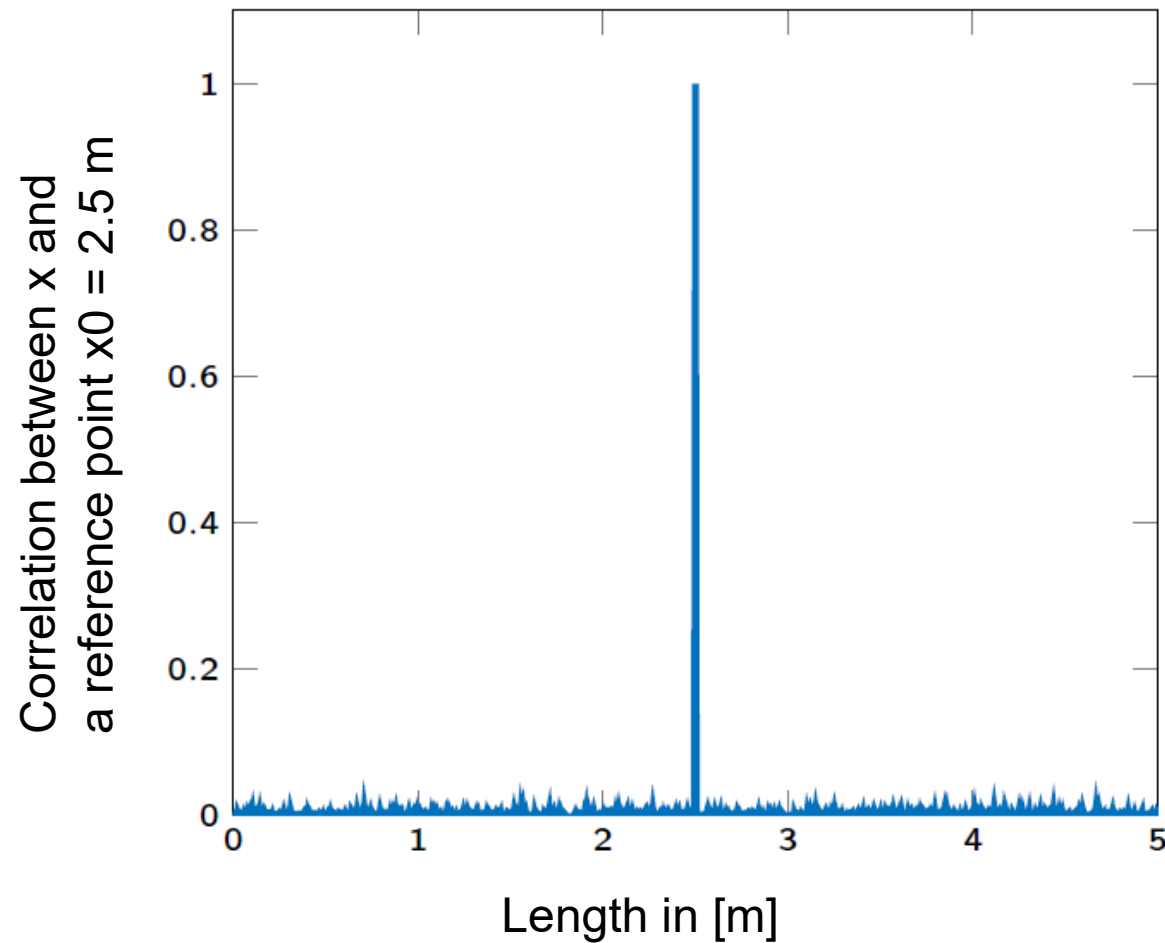
$$C_{HH}(t_1, t_2) = \mathbb{E}\{(H(t_1) - \mathbb{E}\{H(t_1)\})(H(t_2) - \mathbb{E}\{H(t_2)\})\}$$

$$= \int_{\mathcal{D}_x} \int_{\mathcal{D}_y} (x_1 - \mathbb{E}\{H(t_1)\})(x_2 - \mathbb{E}\{H(t_2)\}) \, dx_1 \, dx_2$$

- If  $t_1 = t_2 = t$ ,  $C_{HH}(t, t) = \text{Var}_H(t)$

# Autocovariance Function

100.000 realizations of a Gaussian random field



# Random Field/Process

- **weakly homogeneous:**

$$\begin{cases} \bar{H}(x) &= \text{const.} \\ C_{HH}(x, x + \xi) &= \text{const}(0, \xi). \end{cases} \quad \forall x \in \mathcal{D}$$

where  $\xi \in \mathcal{D}$  defines a vector. When there is no zero entry in  $\xi$  in time, the process is called weakly stationary.

- **strongly homogeneous** in dimension  $d$ : all possible moments are not affected by any shift in dimension  $d$ .

- **isotropic:**

$C_{HH}$  only depends on the distance, not on the direction

$$C_{HH}(x, x + \xi) = \text{const}(\|\xi\|), \quad \xi \in \mathcal{D}, \quad \forall x \in \mathcal{D}$$



# Correlation Length

When  $H$  is a weakly stationary random field with exponential correlation

$$C_{HH}(x_1, x_2) = \exp\left\{-\frac{|x_1 - x_2|}{\ell}\right\}$$

$\ell$  is the correlation length (parameter with the units of  $x$ ).  
It is interpreted as the decay of the correlation between two points of the field.

# Wiener-Khintchine Theorem

If the autocovariance function of a weakly stationary random field holds

$$\lim_{\xi \rightarrow \pm\infty} C_{HH}(x, x + \xi) = 0, \quad \forall x \in \mathcal{D},$$

then, a Fourier transform of  $C_{HH}$  exists.

The spectral density  $S_{HH}(\omega)$  of the weakly stationary field reads

$$S_{HH}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C_{HH}(x) e^{-i\omega x} \, dx$$

# Wiener-Khintchine Theorem

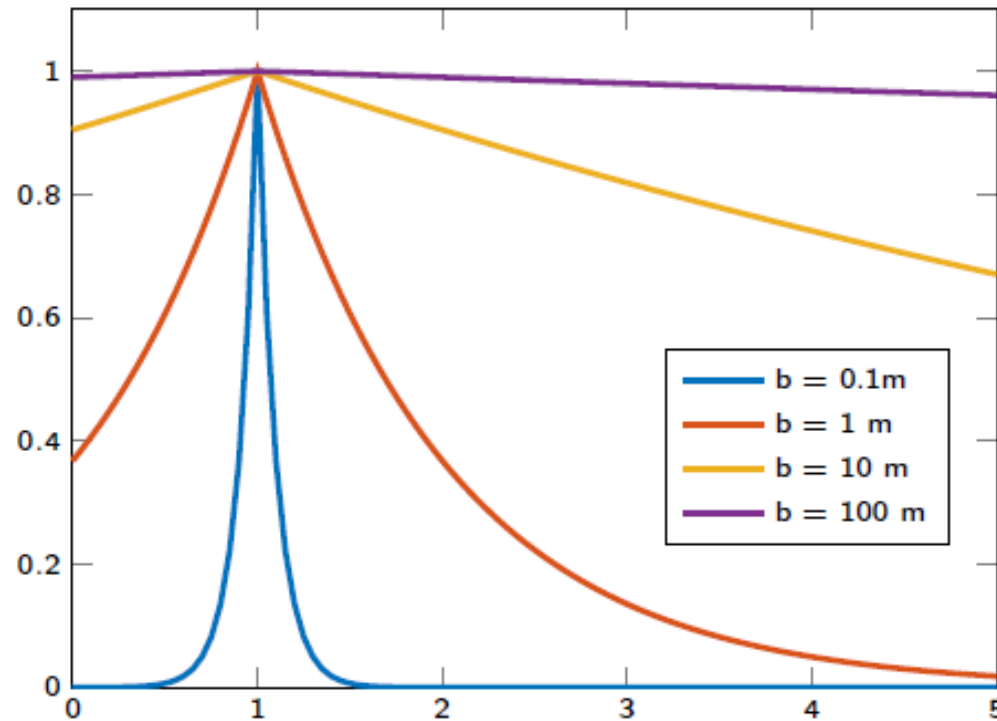
From the spectral density  $S_{HH}(\omega)$  the autocorrelation function is obtained by back-transformation,

$$C_{HH}(x) = \int_{-\infty}^{\infty} S_{HH}(\omega) e^{i\omega x} d\omega$$

Conclusion: The autocorrelation function of a stationary random field can be spectrally decomposed and expressed by the power spectrum of the process.

# Correlation Length

Example: autocorrelation function for a one-dimensional random field for different correlation length at point  $x_0 = 1\text{m}$



# Random Variables - In a Nutshell

## Definition

A random variable  $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$  is a measurable function mapping outcomes to real numbers.

The distribution is fully characterised by  $F_X(x) = \mathbb{P}(X \leq x)$ .

## Key Moments

**Mean (1st moment):**  $\mu = \mathbb{E}[X] = \int x f_X(x) dx$

**Variance (2nd central):**  $\sigma^2 = \mathbb{E}[(X-\mu)^2]$

**Coefficient of Variation:**  $CV = \sigma / \mu$   
(dimensionless scatter)

## Common Distributions in Engineering UQ

**Gaussian**  $N(\mu, \sigma^2)$

*Material stiffness, dimensions*

**Lognormal**  $LN(\mu, \sigma^2)$

*Strength, fatigue life (positive)*

**Uniform**  $U(a, b)$

*Bounded geometric parameters*

**Beta**  $Beta(\alpha, \beta)$

*Fibre volume fraction  $[0, 1]$*

# Sources of Uncertainty

*Aleatoric vs. Epistemic — a fundamental distinction*

## Aleatoric (Irreducible)

*Inherent randomness — cannot be reduced by collecting more data*

- Natural scatter in material properties (grain structure)
- Thermal fluctuations in micro-scale phenomena
- Spatial variability of soil properties
- Live loads: pedestrian, wind, seismic

→ **Modelled as random variables / fields**

## Epistemic (Reducible)

*Uncertainty due to lack of knowledge — reducible with more data*

- Insufficient experimental data for parameter estimation
- Model-form uncertainty (constitutive laws)
- Numerical discretisation error
- Unknown boundary conditions

→ **Modelled via Bayesian inference / interval methods**

*In SFEM: both types can coexist. Proper treatment requires careful problem formulation before choosing a method.*

# Overview on UQ Strategies

*From Monte Carlo to Polynomial Chaos — a first look*

| M4                                               | M5                           | M6                             | M7                                | M8                                  |
|--------------------------------------------------|------------------------------|--------------------------------|-----------------------------------|-------------------------------------|
| <b>Monte Carlo<br/>(MC / MLMC)</b>               | <b>Perturbation<br/>SFEM</b> | <b>Spectral<br/>SFEM (PCE)</b> | <b>Stochastic<br/>Collocation</b> | <b>Surrogates<br/>(Kriging, NN)</b> |
| ✓ General, robust                                | ✓ Fast, low cost             | ✓ Exponential conv.            | ✓ Non-intrusive, flex.            | ✓ Very cheap online                 |
| ✗ Slow convergence<br>( $N^{\{-\frac{1}{2}\}}$ ) | ✗ Small uncertainty only     | ✗ Intrusive, high dim.         | ✗ Curse of dimensionality         | ✗ Training cost, accuracy           |

*Intrusive methods modify the FEM solver. Non-intrusive methods treat it as a black box. Choice depends on cost, accuracy, and dimensionality.*

# Key Takeaways

- 1 **Probability triple  $(\Omega, \mathcal{F}, \mathbb{P})$**  Rigorous foundation: all stochastic analysis starts here.
- 2 **Random variables & distributions** RVs map outcomes to  $\mathbb{R}$ ; characterised by CDF, PDF, moments.
- 3 **Random fields  $H(\mathbf{x}, \omega)$**  Generalise RVs to spatial domains; covariance kernel = key descriptor.
- 4 **Aleatoric vs. epistemic** Fundamental distinction with implications for modelling strategy.
- 5 **UQ strategies overview** MC, Perturbation, PCE, Collocation, Surrogates — each with trade-offs.

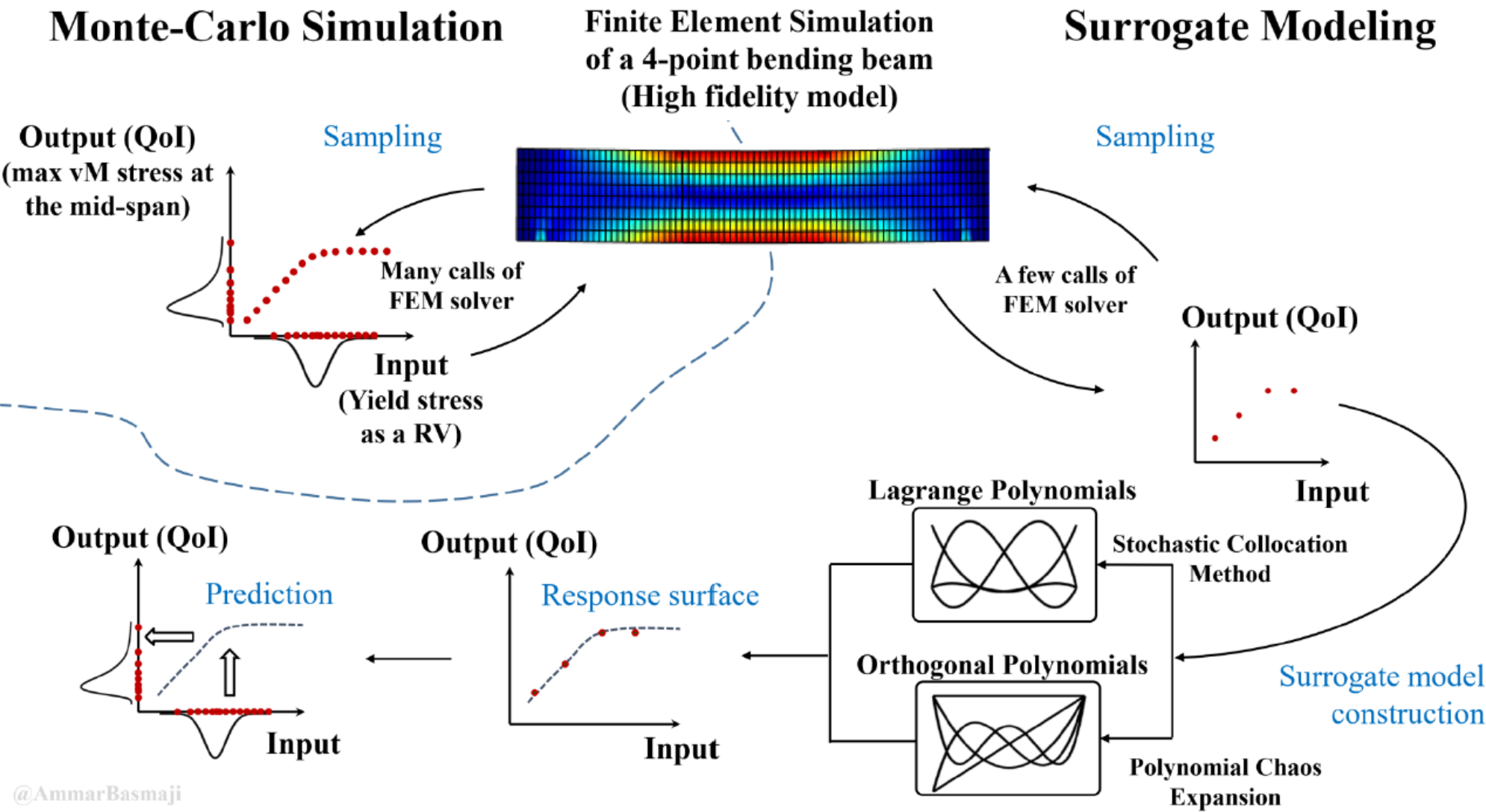
*Next session — Review of Deterministic FEM (weak forms, Galerkin, assembly, error estimates)*



# How to describe Randomness

- One Random Variable (discrete or continuous):
  - Probability density function, cumulative distribution functions,
  - Moments.
- Several Random Variables:
  - Joint cumulative probability density function,
  - Conditional probability density function,
  - Covariance and correlation
- Random Vectors:
  - Covariance matrices.
- Random Fields:
  - Moments,
  - Autocovariance functions and spectral density.

# Procedure



@AmmarBasmaji

# Outline

- Introduction: basics of stochastics and computational mechanics
- Intro to stochastic computations
- Random Fields
- Sampling techniques: Monte Carlo Methods, Quasi MC
- Stochastic Collocation
- Polynomial Chaos Expansion/Sparse PCE
- Spectral stochastic Finite Element Method
- A brief review on reliability analysis
- Modeling of random processes

# Expected Prior Knowledge

- Solid knowledge on Finite Element Techniques and related mathematical basis, i.e. approximate solution of PDE's
- Basic knowledge on Statistics and Stochastic Analysis
- Basic knowledge on modeling of inelastic material behavior and its treatment via FEM (i.e. Mechanics of Solid Bodies)
- Basic knowledge in Matlab (python) programming
- Recommended prior readings:
  - Sudret and Der Kiureghian: Stochastic Finite Element Methods and Reliability, Univ. of California, Berkely, 2000
  - De Souza Neto, Peric, Owen, Computational Methods for Plasticity: Theory and Applications, Wiley 2008
- Recommended prior classes (StudIP/ILIAS):
  - Numerische Mechanik
  - Mechanics of Solids
  - Risk and Reliability Analysis

# Expected Learning Outcome

- Knowledge on available (alternative) methods for uncertainty quantification based on (non-linear) Finite Element computations.
- Ability to work into the state of the art of current methods based on reviewing literature and exploring the needs for future research (investigations).
- Competence to explore a specific technique based on the implementation of simple problems.
- Critical judgement of own results based on the methods of choice etc.
- Scientific writing and presentation skills (report and colloquium).

## Recommended Literature available online via TIB

- Le Maitre and Knio: Spectral Methods for Uncertainty Quantification, Springer, 2010  
<https://doi.org/10.1007/978-90-481-3520-2>
- Ghanem and Spanos: Stochastic Finite Elements – A Spectral Approach, Springer, 1991  
<https://doi.org/10.1007/978-1-4612-3094-6>
- Plaskota and Wozniakowski (eds): Monte Carlo and Quasi Monte Carlo Methods 2010, Springer, 2010  
<https://doi.org/10.1007/978-3-642-27440-4>
- Sudret and Der Kiureghian: Stochastic Finite Element Methods and Reliability, Univ. of California, Berkely, 2000  
<https://ethz.ch/content/dam/ethz/special-interest/baug/ibk/risk-safety-and-uncertainty-dam/publications/reports/SFE-report-Sudret.pdf>

# Examination

- Project and Research Based Learning Approach
- Semester Project, Report and Oral Presentation
- In Detail:
  - Literature work on a specific topic
  - Working out the state of the art and needs for future research
  - Performing some basic computations on that topic
  - Discussion of obtained results
  - Written report, preferably journal style layout, max. 20 pages
  - Oral presentation and discussion

# Recap of FEM

We seek for an approximate numerical solution for the mechanical equilibrium equations (elliptic BVB):

$$\operatorname{div} \boldsymbol{\sigma} + \varrho \mathbf{b} = \mathbf{0}$$

to be solved on a spatial domain  $\mathcal{B}$  with boundary conditions:

$$\begin{aligned} \varphi &= \varphi_0 & \text{on } \partial_\varphi(\mathcal{B}) \\ \mathbf{t} &= \mathbf{t}_0 & \text{on } \partial_t(\mathcal{B}) \end{aligned}$$

We are seeking for a solution in a weighted integral sense (weak form), rather than for a point wise fulfilment of the PDE, e.g. by finite difference approximation of the derivatives,

$$\int_{\mathcal{B}} \delta \varphi \cdot (\operatorname{div} \boldsymbol{\sigma} + \varrho \mathbf{b}) \, dV = 0$$



# Recap of FEM

Integration by parts:

$$\int_{\mathcal{B}} \delta \varphi \cdot \operatorname{div} \boldsymbol{\sigma} \, dV = \int_{\mathcal{B}} \operatorname{div}(\delta \varphi \cdot \boldsymbol{\sigma}) \, dV - \int_{\mathcal{B}} \operatorname{grad} \delta \varphi \cdot \cdot \boldsymbol{\sigma} \, dV$$

Gauss theorem

$$\begin{aligned} \int_{\mathcal{B}} \operatorname{div}(\delta \varphi \cdot \boldsymbol{\sigma}) \, dV &= \int_{\partial_t \mathcal{B}} \delta \varphi \cdot \boldsymbol{\sigma} \cdot \boldsymbol{n} \, dA \\ &= \int_{\partial_t \mathcal{B}} \delta \varphi \cdot \boldsymbol{t}_0 \, dA \end{aligned}$$

Weak form of mechanical equilibrium

$$\int_{\mathcal{B}} \operatorname{grad} \delta \varphi \cdot \cdot \boldsymbol{\sigma} \, dV = \int_{\mathcal{B}} \delta \varphi \varrho \cdot \boldsymbol{b} \, dV + \int_{\partial_t \mathcal{B}} \delta \varphi \cdot \boldsymbol{t}_0 \, dA$$

Introducing constitutive equations, e.g. Hooke's law, in matrix notation

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$$

with the linear strain tensor

$$\boldsymbol{\varepsilon} = \text{sym}(\text{grad}\mathbf{u})$$

yields after applying Gauss theorem the form of mechanical equilibrium

$$\int_{\mathcal{B}} \delta \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} dv - \int_{\mathcal{B}} \varrho \delta \mathbf{u}^T \mathbf{b} dv - \int_{\partial \mathcal{B}} \delta \mathbf{u}^T \mathbf{t} da = 0$$

A lokal ansatz for the displacement field is stated

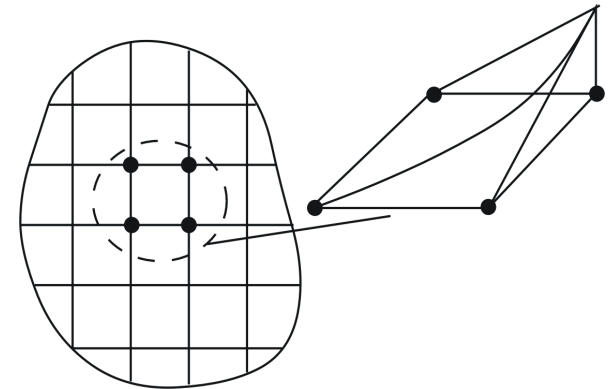
$$\mathbf{u}(\mathbf{x}, t) = \mathbf{H}(\mathbf{x}) \hat{\mathbf{u}}(t)$$

from which the gradient field is approximated

$$\boldsymbol{\varepsilon} = \mathbf{B} \hat{\mathbf{u}}$$

Thus we finally obtain

$$\left[ \int_{V_E} \mathbf{B}^T \mathbf{C} \mathbf{B} dv \right] \hat{\mathbf{u}} - \int_{V_E} \mathbf{H}^T \varrho \mathbf{b} dv - \int_{\partial V_E} \mathbf{H}^T \mathbf{t} da = 0$$



# Assembling and Global Solution

The equilibrium equations obtained element wise, i.e.

$$\mathbf{K}_E \hat{\mathbf{u}}_E = \hat{\mathbf{f}}_E$$

with the element stiffness matrix  $\mathbf{K}_E = \int_{V_E} \mathbf{B}^T \mathbf{C} \mathbf{B} \, dv$

with the element nodal force matrix  $\hat{\mathbf{f}} = \int_{V_E} \mathbf{H}^T \varrho \mathbf{b} \, dv + \int_{\partial V_E} \mathbf{H}^T \mathbf{t} \, da$

are assembled for the discretized domain under consideration of the continuity of the primal variables (equal displacement at common nodes) and the nodal balance equations (equilibrium of forces at common nodes), ending up in the formulation of a global linear system

$$\mathbf{K} \hat{\mathbf{u}} = \hat{\mathbf{f}}$$

to be solved for the unknown nodal displacements with respect to defined Dirichlet and Neumann conditions.

In structural analysis for example the deformation of the structure under mechanical loading is not the major focus of the analyst, rather than the mechanical stresses in order to investigate the load carrying behavior of a structure.

This can be easily done by a post-processing procedure, i.e.:

After the global nodal displacement field has been computed, the gradients can be computed easily on element level by

$$\epsilon = \mathbf{B}\hat{\mathbf{u}}$$

and from that with the constitutive equations (here for linear elasticity)

$$\sigma = \mathbf{C}\epsilon$$

the stress fields are computed (locally inside each element) and typically presented as smooth color maps.

## Remark on the Element Type

There are a plenty of different element formulations available in nowadays powerful commercial (general purpose) codes.

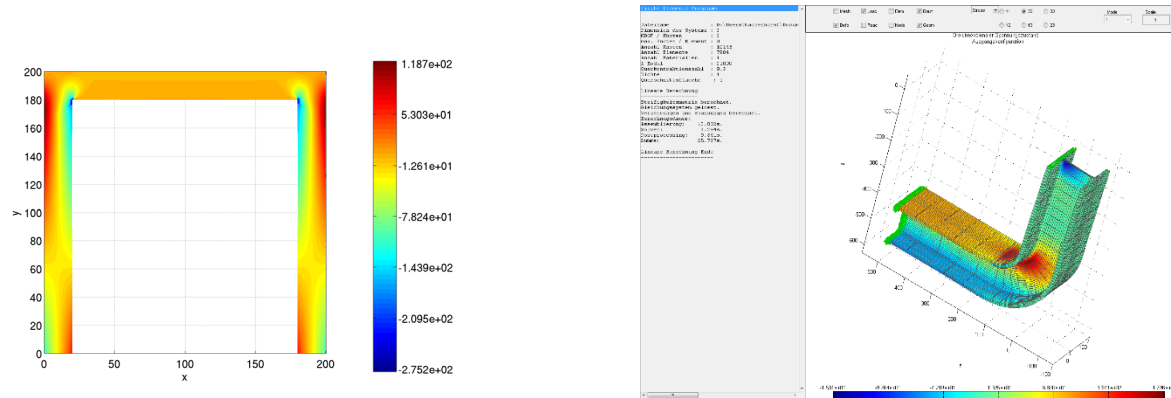
In general, we distinguish between the

- shape of the element and the domains dimension to be discretized, e.g. triangles, quadrilaterals, tetrahedral, bricks etc.
- the order of the polynomial ansatz, e.g. linear, quadratic and higher order ansatz functions
- low order elements with specific enhancements to avoid nonphysical behavior under specific circumstances (e.g. EAS)
- structural elements like beam, plate and shell elements, which require specific mathematical and numerical treatment.

The choice of a suitable element formulation is a responsible modelers tasks and needs to be well documented!

# Remark on the Judgement of Results

Results of a FEA are typically colorful depicted. However, what is the meaning of that under the consideration of model assumptions stated for the specific analysis.



For example in 3D analysis of solid mechanics stress we have to judge 3D deformation maps and 6D stress maps (often compacted to equivalent stresses). At least the plausibility of the analysis should be demonstrated on a simple geometry, where an analytical solution is available.

Before drawing the conclusions of your analysis, all steps with regard to model assumptions, verifications etc. must be well documented.

# Treatment of Nonlinearities

## Classification of Nonlinearities:

- Non-linear and / or time dependent material behavior (material equations are defined locally)
- Geometric nonlinearities, i.e. large deformations
- Non-linear boundary conditions, e.g. contact

## Incremental Solution Strategy:

- Linearization
- Algorithmic treatment

# Linearization

Taylor-series expansion:

$${}^{t+\Delta t}G = {}^tG + \frac{\partial G}{\partial \varphi} \cdot \Delta \varphi$$

Directional derivative (Gateaux):

$$\frac{\partial G}{\partial \varphi} \cdot \Delta \varphi = \left. \frac{\partial}{\partial \epsilon} G({}^t\varphi + \epsilon \Delta \varphi) \right|_{\epsilon=0}$$

Linearized weak form of mechanical equilibrium

$$\begin{aligned} \int_{\mathcal{B}} \text{grad} \delta \varphi \cdot {}^t\mathbf{C} \cdot \text{grad} \Delta \varphi \, dV &= \int_{\mathcal{B}} \varrho \delta \varphi \cdot {}^{t+\Delta t}\mathbf{b} \, dV + \int_{\partial_t \mathcal{B}} \delta \varphi \cdot {}^{t+\Delta t}\mathbf{t}_0 \, dA \\ &\quad - \int_{\mathcal{B}} \text{grad} \delta \varphi \cdot {}^t\boldsymbol{\sigma} \, dV \end{aligned}$$



# Finite Element Discretization

Shape function approach

$$\varphi = \mathbf{H}\hat{\varphi} \quad , \quad \varepsilon = \mathbf{B}\hat{\varphi}$$

Matrix formulation of linearized weak equilibrium equation

$$\int_{\mathcal{B}} \mathbf{B}^T {}^t \mathbf{C} \mathbf{B} \, dV \, \Delta \varphi = \int_{\mathcal{B}} \mathbf{H}^T {}^{\varrho^{t+\Delta t}} \mathbf{b} \, dV + \int_{\partial_t \mathcal{B}} \mathbf{H}^T {}^{t+\Delta t} \mathbf{t}_0 \, dA - \int_{\mathcal{B}} \mathbf{B}^T {}^t \boldsymbol{\sigma} \, dV$$

Incremental finite element system

$${}^{t+\Delta t} G = {}^t \mathbf{K} \Delta \varphi = {}^{t+\Delta t} \mathbf{f}_{ext} - {}^t \mathbf{f}_{int}$$

# Algorithmic Treatment: Newton-Raphson

→ Loop over load increments (counter  $t$ ):

$${}^{t+\Delta t}\mathbf{f}_{ext} = {}^t\mathbf{f}_{ext} + \Delta\mathbf{f} \quad \text{until} \quad {}^{t+\Delta t}\mathbf{f}_{ext} = \mathbf{f}_{max}$$

- linear predictor step:

$${}^t_0\mathbf{K} \Delta_0\varphi = {}^{t+\Delta t}\mathbf{f}_{ext} - {}^t_0\mathbf{f}_{int}$$

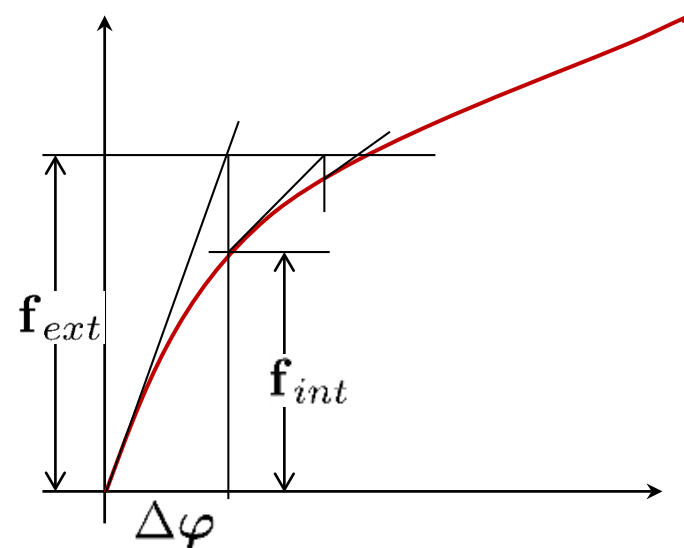
$${}^{t+\Delta t}_1\varphi = {}^t\varphi + \Delta_0\varphi$$

- equilibrium iteration (counter  $n$ )

$${}^t_n\mathbf{K} \Delta_n\varphi = {}^{t+\Delta t}\mathbf{f}_{ext} - {}^t_n\mathbf{f}_{int}$$

$${}^{t+\Delta t}_{n+1}\varphi = {}^{t+\Delta t}_n\varphi + \Delta_n\varphi$$

$$\text{until: } R = \|{}^{t+\Delta t}\mathbf{f}_{ext} - {}^{t+\Delta t}_n\mathbf{f}_{int}\| < \epsilon$$



# Inelastic Constitutive Equations

- We assume a small deformation framework first, i.e. the linear strain tensor appears suitable.
- Material non-linearities are due to in-elastic behavior, which is described by internal variables. Within the thermo-dynamic frame these are formulated as kinetic equations (first order ODE's)

$$\dot{z} = \zeta(z(t), \dots)$$

- Now we are looking for a general framework on solving these equations within a finite element framework.
- Here we have to distinguish between time in-dependent and time dependent processes

# Integration of State Equations

Internal variables are described by evolution equations in time. Mathematically we have to solve single ODE's of 1st order,

$$\dot{z} = \zeta(z(t), \dots)$$

Considering the initial condition

$$z(0) = z_0$$

The state equations for the internal variables describe the inelastic history of the material. They are defined locally, the influence of the surrounding is only captured by the elastic strain (gradient fields).

E.g. numerical solution using an implicit Euler rule

$${}^{n+1}z = {}^nz + {}^{n+1}\zeta \Delta t$$

## Remark on the Modeling Tasks

The analyst must decide and justify the chosen modeling strategy, e.g. 2D vs. 3D, element type, choice of material model, boundary conditions etc. A model verification must well documented and (in ideal) cases a model validation is available.

It does not make sense, that one goes for an expensive stochastic FE analysis, if the model errors are much larger than the variations of parameters!

# Outlook: Stochastic FEM

Assume random material properties and loading

$$\mathbf{K}(\Theta) \hat{\mathbf{u}}(\Theta) = \hat{\mathbf{f}}(\Theta)$$

Solution strategies:

- Sampling the Input Space (MC, QMC)
- Stochastic Collocation Methods
- Perturbation Methods etc.
- Spectral Methods